Donghyun Kim

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Research Experience

Emory Vision and Graphics Lab, Computer Science, Emory University
Visiting Researcher (Advisor: Youngjoong Kwon) Feb, 2026 - Present

- Human Performance Capture
- 3D World Model
- Novel View Synthesis

MICV Lab, Department of Artificial Intelligence, Yonsei University
M.S. / Ph.D. (Advisor: Seong Jae Hwang) Sep, 2023 - Present

- 3D Point Cloud Upsampling
- 3D Point Cloud Representation Learning
- 3D Wireframe Reconstruction
- Text-guided 3D Point Cloud Editing
- Image Dehazing
- CXR Understanding

MICV Lab, Department of Artificial Intelligence, Yonsei University
Under grad. Intern (Advisor: Seong Jae Hwang) Jul, 2022 - Aug, 2023

- Image Dehazing
- Domain Generalization
- Medical Imaging
- Generative Model

DC Lab, Department of Computer Science and Engineering, Yonsei University
Under grad. Intern (Advisor: Kyoungwoo Lee) Jan, 2021 - Jun, 2021

- Project "FluidTrack" with Severance Children's Hospital
  - Develop Tablet PC-based Application
  - Experimenter Interview

YBIGTA, Yonsei University
Data Science Team Member Jan, 2022 - Feb, 2023

- Sessions about CV & ML & NLP & RL
- Study Groups about Generative Model & 3D Point Cloud
- Internal Conferences
  - Project "All-in-Space"
    Sentence Embedding & Clustering
  - Project "Bartender-Y"
    Sentence Embedding & Recommender System

Yonsei AI Exchange Program, Arizona State University
Participant Feb, 2024

Education

Yonsei University Seoul, Korea
B.S. in Computer Science Mar, 2019 - Aug, 2023

Yonsei University Seoul, Korea
M.S. / Ph.D. in Computer Science Sep, 2023 - Present

Papers

PLATYPUS: Progressive Local Surface Estimator for Arbitrary-Scale Point Cloud Upsampling

1st Author AAAI 2025 Oral (*top 4.6%*)

FALCON: Fast Image Haze Removal Leveraging Continuous Density Mask
1st Author CVPR 2025 NTIRE Workshop

How Noisy Poses Break Inverse Dynamics: Analysis and Mitigation for Video-Based Joint Torque Estimation
1st Author CVPR 2026 PhysHuman Workshop

Rethinking Graph Convolution for 2D-to-3D Hand Pose Lifting
2nd Author CVPR 2026 PhysHuman Workshop Oral

Delaunay Canopy: Building Wireframe Reconstruction from Airborne LiDAR Point Clouds via Delaunay Graph
1st Author under review

Fourier Decomposition for Explicit Representation of Colored Point Cloud Attributes
1st Author under review

WoLF: Wide-scope Large Language Model Framework for CXR Understanding
3rd Author under review

Professional Experience

Program Committee
ECCV 2026 MoCha Workshop

Conference Reviewer
WACV 2024, WACV 2025, ICCV 2025, CVPR 2025 NTIRE Workshop, 3DV 2026, CVPR 2026, ECCV 2026, NeurIPS 2026

Journal Reviewer
Pattern Recognition, Transactions on Image Processing

Teaching Assistant
Introduction to Deep Learning and Application, Yonsei University Mar, 2024 - Jun, 2024
Introduction to Deep Learning and Application, Yonsei University Mar, 2025 - Jun, 2025
Introduction to Deep Learning and Application, Yonsei University Mar, 2026 - Jun, 2026

Lecture Instructor
DX (Digital Transformation) Training, Samsung Electronics Sep, 2023 - Oct, 2023

Industry Experience

Linz Private Equity
Intern Jan, 2020 - Feb, 2020

- Stock Data Analysis

Innoyard
Software Engineer, Researcher Jun, 2021 - Aug, 2022

- Project "InnoScan" - Mobile 3D Human Body Data Capturing Service
  - Develop Mobile & Tablet PC-based Application (Flutter & Swift)
  - Collect 3D Point Cloud Data
  - Research on 3D Data Processing (Point Cloud Registration & Surface Reconstruction)

Awards & Honors

Coding Challenge, Korea Institute of Startup & Entrepreneurship Development(KISED)
2nd Prize Dec, 2021

Software Capstone Project, Department of Computer Science and Engineering, Yonsei University
Exhibition Popularity Award 2nd Semester, 2022

Software Capstone Project, Department of Computer Science and Engineering, Yonsei University
Grand Prize 1st Semester, 2023